

DREAMS, RESILIENCE AND MAKING A DIFFERENCE

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Beta Phi Mu Theta Initiation
Pratt Institute, School of Information
May 16, 2017**

MAKING A DIFFERENCE



Manuscripts and Archives Division, The New York Public Library. Work with schools : a 'private' lesson, child showing something in book to librarian, ca. 1920?. Retrieved from <http://digitalcollections.nypl.org/items/510d47da-e5fd-a3d9-e040-e00a18064a99>

DREAMS



Antonio de Pereda. (ca 1650). The Knight's Dream. Retrieved from https://commons.wikimedia.org/wiki/File:Antonio_de_Pereda_-_El_sue%C3%B1o_del_caballero_-_Google_Art_Project.jpg

RESILIENCE



FEMA/Joselyne Augustino. (July 23, 2013). A home in Gilchrist, Texas, designed to resist flood waters survived Hurricane Ike in 2008. Retrieved from https://commons.wikimedia.org/wiki/File:Home_designed_to_resist_flood_waters.jpg

A close-up photograph of a disorganized pile of papers, books, and documents. A large mass of white packing peanuts is visible, partially covering the items. The scene is dimly lit, emphasizing the clutter.

SOMETIMES IT LOOKS
LIKE THIS...

A photograph of a metal shelving unit filled with numerous index cards. The cards are neatly organized into rows on each shelf, creating a structured and orderly appearance. The lighting is bright, highlighting the uniformity of the arrangement.

WHEN WE WANT IT TO
LOOK LIKE THIS...

RESILIENCE

re·sil·ience

/rəˈzilyəns/

noun

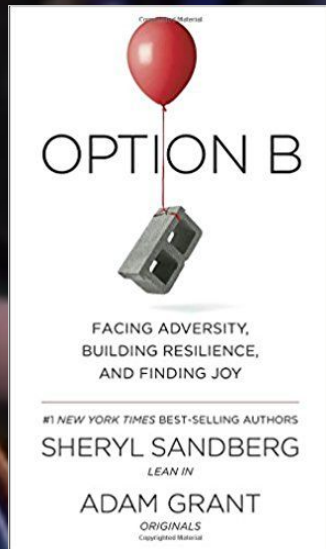
1. the capacity to recover quickly from difficulties; toughness.
"the often remarkable resilience of so many British institutions"
2. the ability of a substance or object to spring back into shape; elasticity.
"nylon is excellent in wearability and resilience"



Translations, word origin, and more definitions

"TO BE RESILIENT, DON'T BE TOO VIRTUOUS"

Grant, Adam. (May 13, 2013). To Be Resilient, Don't Be Too Virtuous. [Commencement Speech]. Retrieved from <https://www.linkedin.com/pulse/resilient-dont-too-virtuous-adam-grant>



Adam Grant's Commencement Speech at Utah State University, May 2017

RESILIENCE IS THE KEY TO:

GENEROSITY

Being generous
on the days when
you lose faith
in humanity

AUTHENTICITY

Staying true to
yourself on the
days when others
lose faith in you

GRIT

Persevering on
the days when
you lose faith
in yourself

RESILIENCE IS MORE THAN GRIT

NYC Office of Emergency Management. (2013). OEM Biennial Report 2013. Retrieved from http://www1.nyc.gov/assets/em/biennial2013/index_1.html#28

PERSONAL RESILIENCE

SECURITY

MINDFULNESS

PLANNING

U.S. Navy photo by Photographer's Mate 2nd Class Jim Watson. (Sept. 14, 2001). A New York City firefighter attempts to clear his eyes of soot during rescue efforts at the World Trade Center. The building was destroyed during a Sep. 11, 2001, terrorist attack. Retrieved from https://commons.m.wikimedia.org/wiki/File:September_14_2001_Ground_Zero_03.jpg #mw-iump-to-license



INSTITUTIONAL RESILIENCE

A photograph showing firefighters in full protective gear working amidst the massive pile of rubble and twisted metal at the World Trade Center site on September 11, 2001. In the background, the skeletal remains of the towers are visible against a hazy sky. The scene is one of devastation and active emergency response.

SECURITY

TRANSPARENCY

PLANNING

CNN. (September 11, 2001). Firefighters, :

A STORY OF RESILIENCE



CNN. (September 11, 2001). Firefighters, :

9/11 RESPONSE



Don Halesy, Library of Congress. (September 11, 2001). Man covered with ashes assisting a woman walking and holding a particle mask to her face, following the September 11th terrorist attack on the World Trade Center, New York City

NOLA: HURRICANE KATRINA



Left: DailyKos, various.. (n.d.). The Water is Rising.
Right: DailyKos, various.(n.d.). Flood, Orleans Parish.

TOHOKU TSUNAMI, MINAMISANRIKU, JAPAN



Shinichi Sato/AP, Yahoo! News. (March 11, 2011). Minamisanriku Government Disaster Readiness Center. Tohoku Tsunami.

EMERGENCY MANAGEMENT IN NYC

9/11 as a catalyst for cooperation

- NYCMAP
- Pier 92: Emergency Mapping and Data Center
- Office of Emergency Management

Public engagement

- Big Apps Competition
- Open Data Mandate
- Road Map for the Digital City

Deep Infrastructure Group

GeoSymposium 2001-2011-2021

NYC Geospatial Catalysts

NYC Geospatial Technology Center

INNOVATIONS IN EMERGENCY RESPONSE SYSTEMS

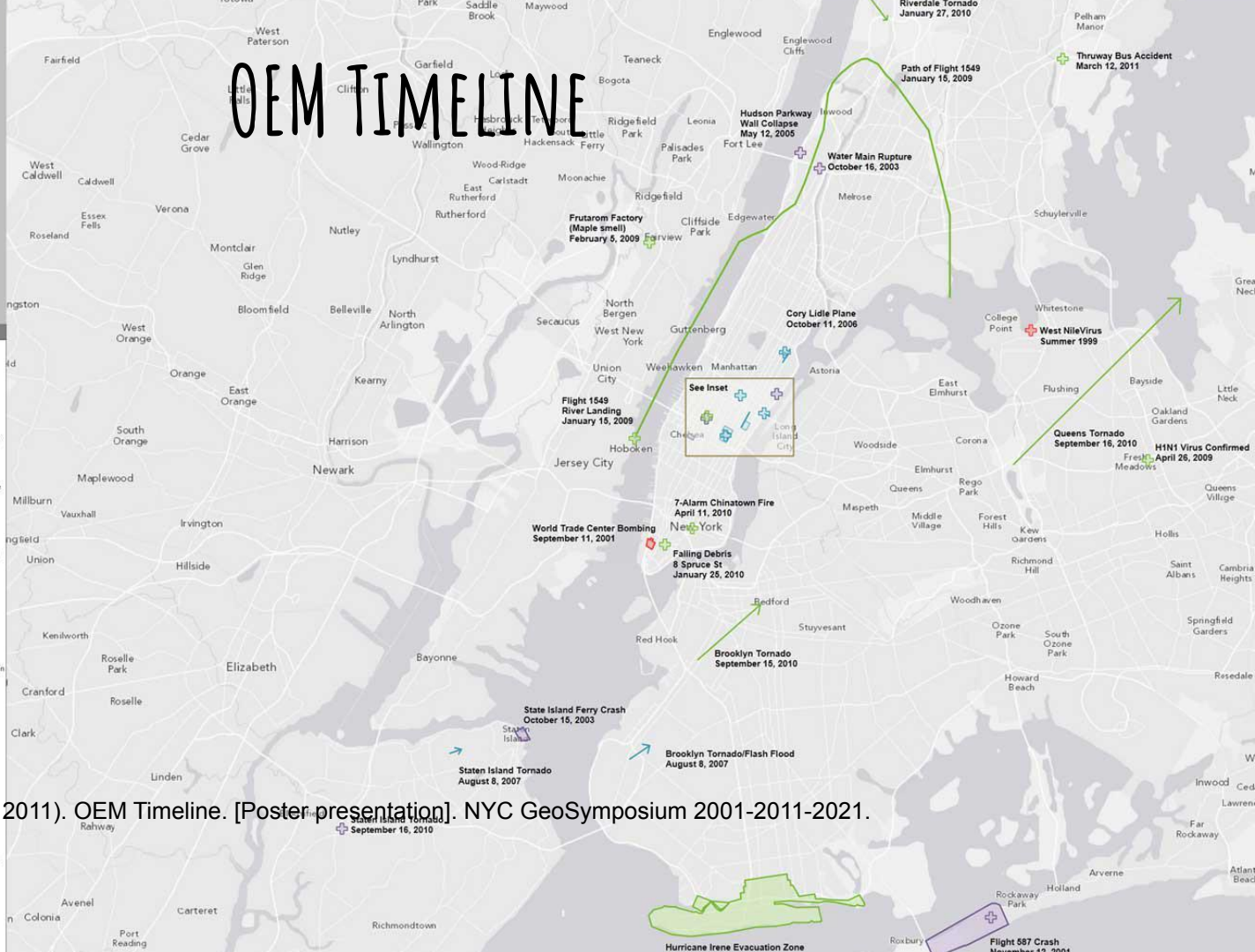
- Period One - Pre-9/11: Baseline
- Period Two - 2001-2005: Characterized by development of information systems to support interagency cooperation and response management.
- Period Three - 2006-2008: Characterized by continued integration and interoperability of information systems across agencies.
- Period Four - 2009-2011: Characterized by improved training and simulation software, and open public initiatives.

Midtown Incidents



Pre-2001	Event	Date of Event	Date Introduced/Technology
	West Nile Virus confirmed	Summer 1999	1/2000 DoITT GIS Unit Established
9/11/2001	9/11 Terrorist Attack	9/11/2001	
2001	Flight 587 Rockaway Plane Crash	11/12/2001	
2003	Staten Island Refinery Fire	2/21/2003	2001 UVIS: Unified Victim Identification 11/2001 OEM Established as a City Department 2003 Mobile Data Center (GIS bus)
	East Coast Blackout	8/14/2003	3/11/2003 NYC 311 Call Center Opens
	Staten Island Ferry Crash	10/15/2003	7/10/2003 Ready New York: Household Emergency
	Washington Heights Water Main Rupture	10/16/2003	
2004	Staten Island Tornado	10/27/2003	11/1/2003 CERT: Community Emergency Response 1/2004 CALMS: Citywide Asset and Logistics Management System
	Republican National Convention	8/30/2004	2004 UVIS: Unified Victim Identification 2004 NYC Map Portal Launched
2005	Henry Hudson Parkway Wall Collapse	5/12/2005	2005 UVIS: Unified Victim Identification System Family Assistance Module
	Thanksgiving Day Parade Balloon Hits	11/25/2005	5/11/2005 Citywide Incident Management System 5/2005 Local Law 47 (public, monthly 311)
2006	Roosevelt Island Tram Rescue	4/19/2006	7/2005 OEM Website Redesign 2006 Interagency Command Vehicle
	Gas explosion and building collapse	7/10/2006	5/12/2006 2nd CERT Training Class
	Heat Emergency	7/14/2006	
	Heat Emergency	7/31/2006	
	Cory Lidie Plane Crashes	10/11/2006	11/30/2006 NYCMap 1.0 Launched 12/5/2006 OEM Headquarters Opens at 165 Cadman 2007 SAHANA Shelter Management System implemented
2007	Steam Pipe Explosion, Midtown Brooklyn Tornado/Flash Flood	7/18/2007 8/6/2007	10/2007 Continuity of Operations Planning 2008 UVIS: Unified Victim Identification System Pandemic Influenza module
2008	Crane Collapse in Turtle Bay	3/15/2008	2/14/2008 NYCStat/CPR Launched
	Pipe Benedict XVI Visit	4/18/2008	
	Crane Collapse	6/19/2008	
2009	US Airways 320XLR Near Midtown	1/19/2009	3/18/2009 Advanced Disaster Management 6/30/2009 YouTube Ready NY Launch
	Frutaron Maple Syrup Smell	2/5/2009	2009 CIMS Mobile Coordination Center 2009 CIMS Learning Management System
	H1N1 Virus Confirmed in NYC	4/26/2009	4/14/2009 NYCMap 2.0 Launched 5/19/2009 NYCWIN-NYC Wireless Network TVS - Interoperable Video Management
	Helicopter/Plane Collision	8/8/2009	5/28/2009 Notify NYC Public Launch 6/16/2009 OEM Facebook Page Launched 6/29/2009 311 Online Launched along with other
2010	High Wind Warning	8/17/2009 1/25/2010	1/14/2010 Notify NYC adds School reporting
	Winter Storm	2/25/2010	
	Chinatown 7-Alarm Fire	4/11/2010	4/28/2010 Notify NYC adds Missing Persons
	Times Square Revolution Attempt	5/1/2010	

OEM TIMELINE



Whysel, Noreen. (November 16, 2011). OEM Timeline. [Poster presentation]. NYC GeoSymposium 2001-2011-2021.

World Trade Center Bombing September 11, 2001

OEM TIMELINE

World Trade Center 2001

Agencies: OEM, Mayor, FDNY, NYPD, USACE, FEMA, ARC, others

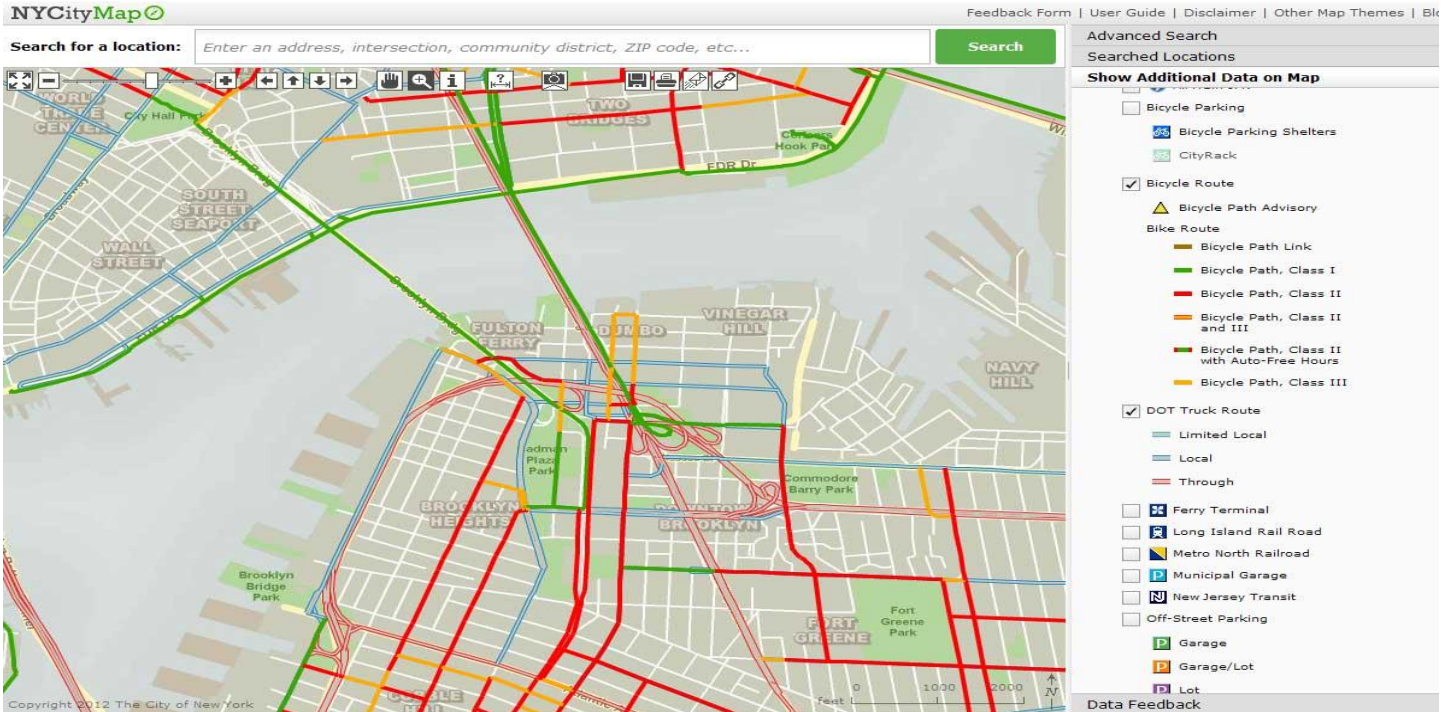
Location: World Trade Center, 16 block environs

With the Mayor's Office of Emergency Management headquarters, sited in 7 World Trade Center, evacuated, an improvised Emergency Operations Center was established on Pier 92, located on the Hudson River at West 52nd Street. With no access to OEM resources at 7 WTC, this event represented the world's largest, ad hoc response situation, and involved the assistance of local and international teams of first response units, public and private volunteers and aid organizations.

[Zoom to](#)

Whysel, Noreen. (November 16, 2011). OEM Timeline. [Presentation]. NYC GeoSymposium 2001-2011-2021. Retrieved from <http://explorer.arcgis.com/apps/presentation/index.html?webmap=06dce9c9c5734e6dbb54fe8c8d6e352c>

NYCITYMAP (2006)

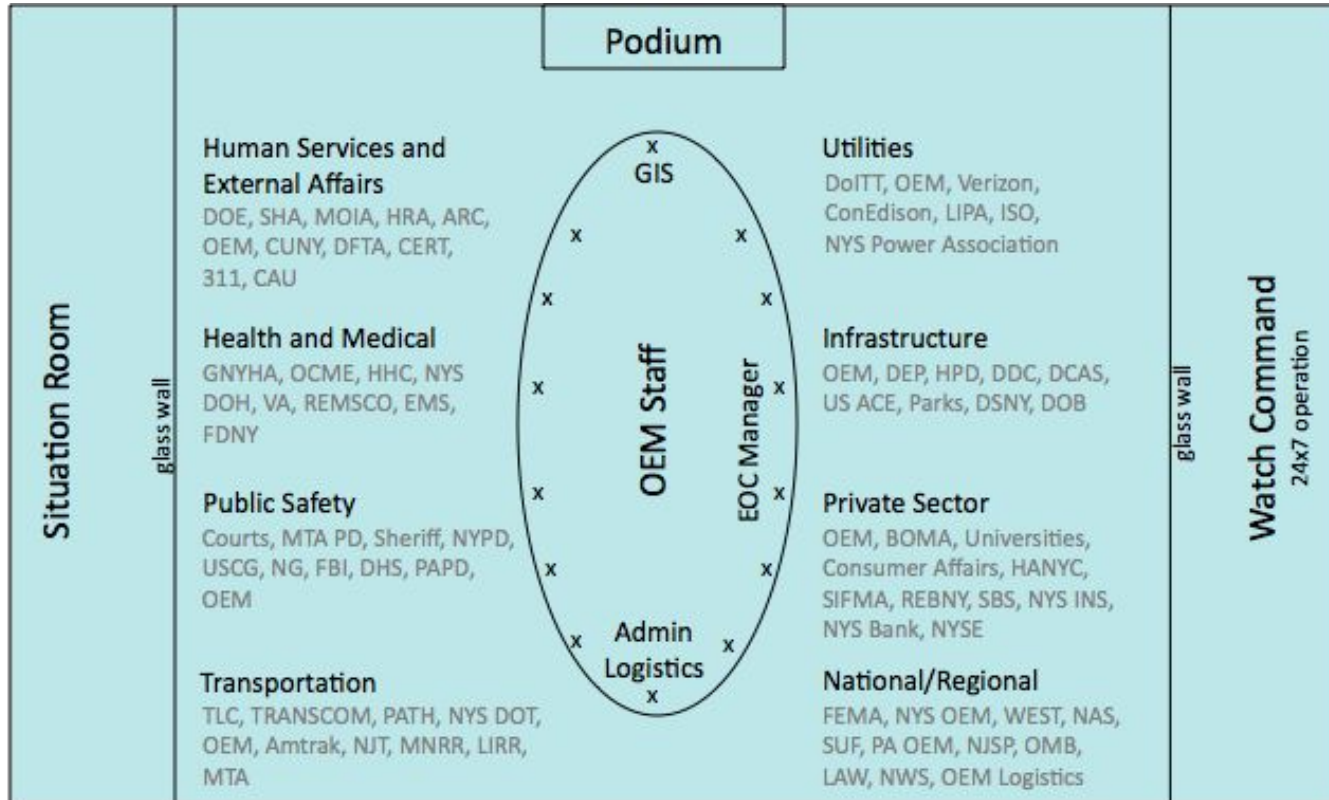


<http://gis.nyc.gov/doitt/nycitymap/>

HURRICANE EVACUATION ZONE FINDER (2006)



EMERGENCY OPERATIONS CENTER PLAN



x = OEM staff

OEM EMERGENCY FACILITIES



Emergency Operations Center

FDNY EMERGENCY FACILITIES



NYC INFORMATION POLICY

- Providing citizen access to data
- Promoting NYC as a technology sector
- Data initiatives modeled after citizen efforts and programs in other cities:
 - Big Apps Contest
 - Change By Us NYC
 - City Council Open Government Mandate

“If we’re going to continue leading the country in innovation and transparency, we’re going to have to make sure that all New Yorkers have access to the data that drives our city.” 2012 Press Release, NYC Mayor’s Office

OPEN DATA INITIATIVES



APP CONTESTS AND HACKATHONS

CODE *for*
AMERICA

**REINVENT
NYC.GOV**



BIGAPPS CONTEST



[About](#) [Challenges](#) [Partners](#) [Workshops](#) [Blog](#)



Read more about this year's
competition and get access to our
workshop resources on our blog!

[Read the Blog](#)



NYC BigApps is the premier civic innovation competition in New York City that challenges designers, developers, academics, entrepreneurs, and New Yorkers at large to apply their know-how to improve New York City.

NOTIFY NYC

3+ Alarm Fire	Electrical	Road Closure - Planned
Accident	Evacuation	Road Closure - Unplanned
Accident/Emergency	Ferry Disruption	Senior Found
Aerial	Fireworks	Severe Thunderstorm
Air Quality	Flood	Significant Event
Aircraft & Egress	Gas Main	Simulated Activity
Aircraft Only	Gas Release / Leak	Simulated Fire
Airport Disruption	H1N1	Simulated Fire/ Explosives
Alert Cancelled	HAZMAT Condition	Snow
Alternate Side Parking	Heat	Struck
Beach Status Change	Heat & Air Quality	Structural Collapse
Blasting / Demolition	Hurricane	Survey
Bridge /Tunnel Closure	Informational	Termination
Bridge Closure	Issuance (mostly missing persons)	Tornado
Brush Fire	Mass Transit Disruption	Tornado Watch
Ceremony / Gun Salute	Notification	Tropical Storm
Child Found	Oil Spill	Water Main
Closure	Other	Weather
Closure/Citiwide	Outside of NYC	Winter Storm
Collapse	Phone	WNV Aerial Spraying
Disruption - Other	Power Outage	WNV Ground Spraying
Earthquake	Relocation	World Trade Center



SOCIAL MEDIA

The top section shows the Facebook profile of the NYC Office of Emergency Management. The profile picture is the OEM logo. The cover photo shows a group of people in a meeting. The page has a 'Wall' tab selected, showing a post from the OEM about the NY 9/11 story highlights. The post includes a photo of a school drill and a link to 'NY 1 Reports: Japanese Students Credit Drills In Tsunami Survival'. Below the post are 'Like', 'Comment', and 'Share' buttons, and a comment from 'Bev Corvin'.

The bottom section shows a YouTube video player. The video title is 'Ready New York Video with St. John's University'. The video player shows a yellow background with the text 'READY NEW YORK' and 'NYC OFFICE OF EMERGENCY MANAGEMENT' below it. The video player controls show a play button, a progress bar at 0:00 / 1:04, and a volume icon.

The screenshot shows a Twitter feed with seven tweets from the account NYC OEM (@NotifyNYC). The tweets are dated 14 Mar. The tweets are as follows:

- NYC OEM @NotifyNYC** 14 Mar
The issued AMBER Alert on 3/14/12 at 8:20 PM has now been canceled.
- NYC OEM @NotifyNYC** 14 Mar
AMBER Alert Navita Jagdeo, age 7 Months, from Queens General Hospital. Please contact 911 if seen bit.ly/w6YXsh
- NYC OEM @NotifyNYC** 14 Mar
Uptown 4 train is running local from BK Bridge to 42nd St and uptown 5 train is running on the 2 line from Nevins St to 149th St.
- NYC OEM @NotifyNYC** 14 Mar
Silver Alert for Wilfred Morris M/B/70: 5'5" tall/140lbs/blue hat/gray hoodie/tan pants/glasses. If seen call 911. yfrog.com/oddnfjhj
[View photo](#)
- NYC OEM @NotifyNYC** 14 Mar
Following a previous motor vehicle accident, the westbound BQE has been reopened at Northern Boulevard. Expect residual delays.
- NYC OEM @NotifyNYC** 14 Mar
Due to a motor vehicle accident, the westbound BQE is closed at Northern Boulevard. Expect delays in both directions.
- NYC OEM @NotifyNYC** 14 Mar
Following an earlier police incident, normal service has resumed on the northbound 1 train. Expect residual delays. mta.info
- NYC OEM @NotifyNYC** 14 Mar
Due to police activity at W28th Street in MN, the n/b 1 train is running express from Chambers St to Times Sq. mta.info.

BACK TO DREAMS



illustrators of the 1890 Holman Bible. (1890). Josephs Dream, as in Genesis 37:9–10, Retrieved from
https://commons.wikimedia.org/wiki/File:Holman_Josephs_Dream.jpg



9/11 & Geoinfrastructure Resources



nwhysel@gmail.com

File Edit View Insert Format Data Tools Add-ons Help All changes saved in Drive

Comments

Share



fx

OEM Interviews TC Hour 3

	A	C	D	E	F	G	H	
1	Title	Date	Creator	Contributors	Description	Date Digital	Rights	Lo
2	OEM Interviews TC Hour 3	2001	Bruce Cahan, Urban Logic, Inc.			na	Video resource (i Ne	
3	OEM Interviews TC Hour 4	2001	Bruce Cahan, Urban Logic, Inc.			na	Video resource (i Ne	
4	OEM B-Roll TC Hour 5	2001	Bruce Cahan, Urban Logic, Inc.			na	Video resource (i Ne	
5	October 16th Shoot Tape 1	10/16/2001	Bruce Cahan, Urban Logic, Inc.			na	Video resource (i Ne	
6	October 16th Shoot Tape 2	10/16/2001	Bruce Cahan, Urban Logic, Inc.			na	Video resource (i Ne	
7	Innovations in American Government: 2000 Finalists Presentations	2000	John F. Kennedy School of Government, Harvard Un				na	Copyright 2000, Har
8								
9	WTC Images: BEFORE, 9/13, 9/15, 9/17, 9/19, 9/22, 9/26, 9/30, 10	10/01/2001, 10/0	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
10	Aerial Photo of WTC & Immediate Area (In WTC Images)	10/01/2001 2:00	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
11	Ground Zero - 09/13/01 (World Trade Center Conditions Morning c	10/01/2001 2:16	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
12	Ground Zero - 09/15/01 (In WTC Images)	10/01/2001 2:25	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
13	Ground Zero - 09/17/01 (In WTC Images)	10/01/2001 2:27	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
14	Ground Zero - 09/19/01 (In WTC Images)	10/01/2001 2:35	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
15	Ground Zero - 09/22/01 (In WTC Images)	10/01/2001 2:42	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
16	Ground Zero - 09/26/01 (In WTC Images)	10/01/2001 2:47	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
17	Ground Zero - 09/30/01 (In WTC Images)	10/01/2001 12:48	Emergency Mapping Center, City c			Images of the W	10/01/2001	Copyright 2001 C Ne
18	Ground Zero - 10/03/01 (In WTC Images)	10/01/2001 1:52	Emergency Mapping Center, City c			Images of the W	10/03/2001	Copyright 2001 C Ne
19								
20	FGDC '02 GISMO '04	2002, 2004						
21	FGDC Presentation	7/19/2002 10:48	Federal Geographic Data Committ			Presentation: inc	7/19/2002 10:48	Copyright 2002, Ne
22	Overview of Process Used to Geocode Claims, June 25, 2002	6/28/2002 1:11 P	NYC GIS Utility	Alan Leidner	Overview of Proc	6/28/2002 1:11 P	Copyright 2002, Ne	



9/11 Items

Publications

Externally Held Resources

LCSH

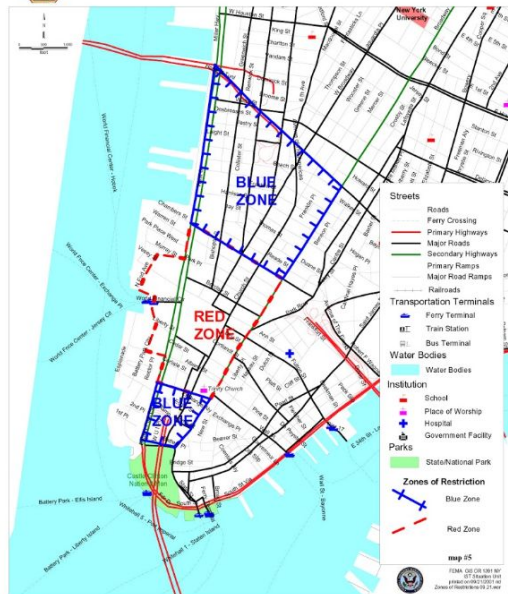
Macromedia Player Files



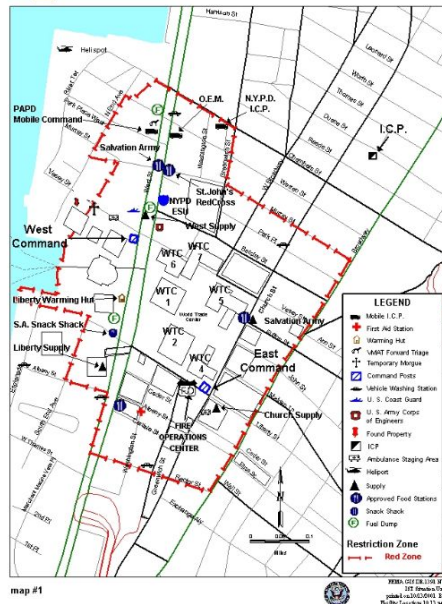
Explore



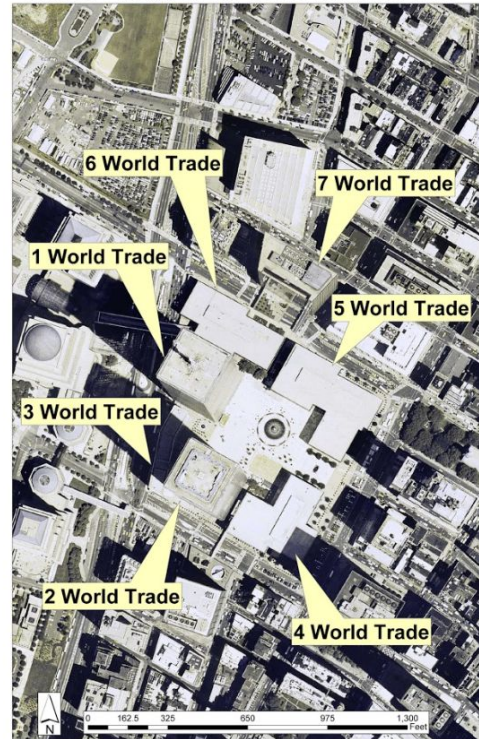
Zones of Restriction
Indicator Map
(as of 2030, September 21, 2001)



DR-1391-NY New York City
Facility Locations - October 13



Aerial Photo WTC & Immediate Area



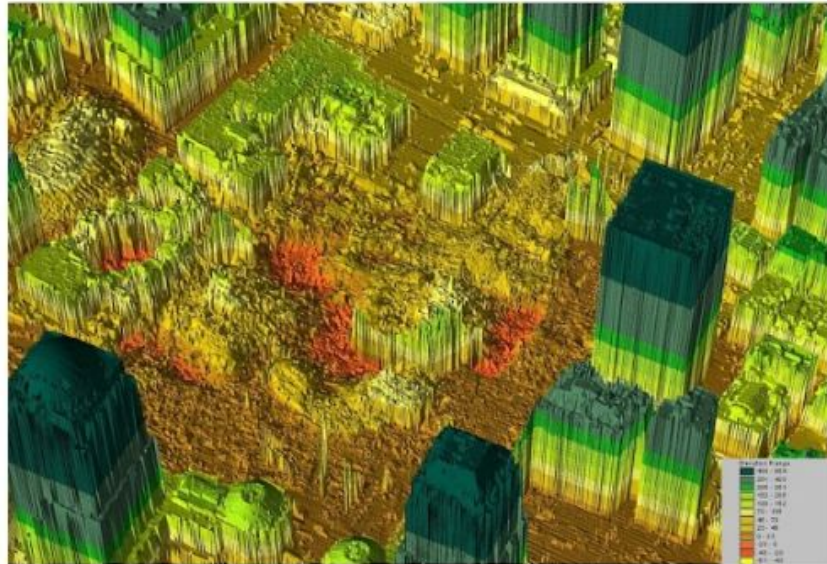
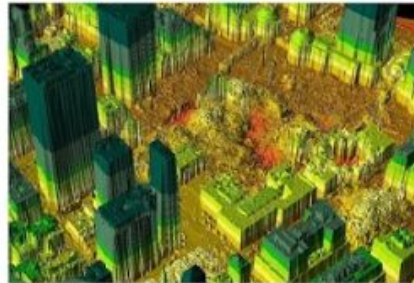
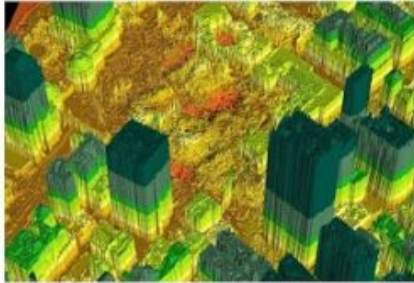
City of New York
Emergency Mapping Center
Rudolph W. Giuliani, Mayor

Disclaimer: The City of New York does not certify the correctness of any information provided on this map and is not liable for any actions taken or not taken by any persons.

LIDAR Images of World Trade Site

Source: USGS LIDAR Imagery 9/17/2001

Emergency Mapping and Data Center
9/18/2001 9:15pm
File: LIDAR Images.psd





BUILDING STRUCTURAL STATUS as of 09/21/01 8:00PM

- Not Affected
- Inspected But Not Affected
- Damaged But Stable Repairs/Cleaning Required
- Major Structural Damage No Occupancy
- Partially Collapsed
- Destroyed



ing and Data Center

GIS

POLICE
U.S. AGENT





MEETING NOTES

UPDATE BY 4PM - Time Stamped Versions

- 1 Utilities Status ^{Conrad} 2PM Doug/Hall 8:15 AM & large
- 2 * Pedestrian Access ^{Vehicle} Technical / Sgt Parsons Doug/Paul/K
- 3 Major Crossings S of 14th (Doug) - In Deal OCT - DATE NOT YET
- 4 Train Status ^{Water} // Bus Map ^{new} EBT
- 5 Buildings Status Hunter Sean 6:30 PM 5:15 PM Blinn
- 6 Red Cross Locations? Mike (Kavanaugh) check out (phone)
- 7 Imagery ^{Halt Office} State Office ^{Thurman} 6:00 AM 3:00 PM CD's to CD case Cherry Pick Imagery Hunter Process EBT procedure

OEM request: Color copies — get

CD for everyone - chronology

CD first of them

* Scanning Service - *
Contractor document Company B & W Buildings

→ Open For Business?

Sept 24, 2011

Postoperative QRS - AM

[illegible]

Wendy Duff

7. Globalisation URS / DEEP INFRASTRUCTURE

Marina Havan Plan Graphics

~~X~~ IBOR Simon
George Davis

X ROBERT NICHOLSON, PE DAVIS ASSOCIATES, INC

? John Pantazis
Urban data Solutions

Ronald Matzner
FGDC/I-Teams

Bruce Cahan UrbanLogic

Barry Blanchard Intergraph Corp

David Armstrong Intergraph Corp.

Rob PATTEN Intergerin Gel.

Jean To PlanGraphics, Inc.
CHRIS SCHIEK ESQ

Russ Johnson
2-1000 HIGHT
10-100000

2-1000 HAIT
10.10000.

CHRONOLOGY

September 11...

GIS Staffing

- FDNY – 3
- NYC – 7
- NYS – 8
- FEMA – 1

FDNY

- Creating products using street centerlines, block outlines, and pre-event imagery
- Producing products immediately
- Reports that they expect that there will be no power in the area for next 60 to 90 days

NYC

- Emergency Operations Center (EOC) evacuated immediately
- EOC Collapses at 6:00pm
- Emergency Operations established @ NYPD Academy
- Meeting to establish GIS support

- New EOC brought CDs containing the NYCMAP
- Producing as-was products at Hunter College at 6:30 pm, delivered at 10:00 pm to police academy
- Products requested/required by the Mayor for the duration
- Planning to relocate the EOC & GIS to a new location
- There is a desperate need for current, high-resolution air photos, but “no-fly zone” precludes civilian air photo missions

NYS

- State Emergency Management Office (SEMO) is Activated immediately
- Office of Technology (OFT) sends 3 staff to SEMO with 3 laptop computers
- Producing closure and critical facilities products
- OFT discussing the need for imagery

September 11 Cont.

FEMA

- Region 2 moves to an alternate facility with one GIS laptop
- Region 1 Regional Operation Center (ROC) Supporting Region 2
- Region 9 ROC activated as an additional backup
- All 10 ROCs activated
- Headquarters Support Available 24x7
- President Bush signs Disaster Declaration for the state of New York
- Imagery and Vector data posted to the internet by ESRI

Additional activities

- SPOT Data Capture

LESSONS LEARNED

Lessons Learned...

- Co-location of GIS assets, a “Bee Hive”, makes for a good working environment in disaster response
- Information Overload comes on quickly, be aggressive with the coordination of data collection and acquisition
- Duplication happens. Minimize it!
- Incident Data Sharing & Warehousing is a must.
- Remote Sensing Data needs to be available 6-12 hour after acquisition to be valuable
- Contacts must be identified within the New Emergency Management Structure created when multiple agencies co-locate
- “Mission oriented” mindset transcended institutional barriers, facilitating data sharing and minimal redundancies. People want to help!
- Security and confidentiality agreements may be needed to ensure data development and sharing by private entities.
- Most requesters still require paper maps
- Inter-Governmental relations need to be nurtured
- Maintain data contacts both inside and outside your area of concern
- Data Collection Standards should be established pre-event and reviewed prior to any post-event collection. It needs to be recognized that data is not being collected for a single purpose
- Network and telecommunications are not guaranteed. When the landlines went down, we were dependent on mobile phones. Internet connectivity was unstable in the first couple weeks.

Lessons Learned Cont.

- Map product standards help identify the vintage and source of products
 - Colors, Symbols, Paper Sizes, Title
 - Product Tag with office, date, and project name
- A database Manager is crucial in large operations

COLLABORATIVE EFFORT

GIS Participants through October 12, 2001

City Agencies

DCP
DEP
DOF
DOH
DoITT
DOT
FDNY
HPD
Landmarks
NYPD
OEM
NYCHA
NYPIRG-CMAP
Parks

State Agencies

NYS DEC
NYS DOT
NYS DOH
MTA / NYCT
MTA / LIRR
NYS OFT
SEMO
NY State Police

Federal Agencies

US DOD
USEPA
FEMA
NASA
NIMA
USACE
USFS

Other Government
CA OES
PA NY/NJ
MN DNR
NPS Midwest Region
Rockland County
UT BLM
Westchester County

Utilities

Con-Ed
Empire City Subway
Verizon

Consultants

Davis Associates
ESRI
Hunter College
MapInfo
MITRE Corp
Plangraphics
Urban Logic
URS

Vendors

Compaq
Dell
EarthData
Hewlett-Packard
IBM
Intel
Keystone(imagery)
LinksPoint
Microsoft
Pictometry Inc.
Trimble

Volunteers

AKRF
American Museum of Natural History
Applied Geography
Baker Engineering
Columbia University
Community Cartography
GISMO
GreenMap
LIRR
NYPIRG/CMAP
Parsons Brinkerhoff
SpaceTrack
Teqical Services Inc (TSI)
United Nations
Wildlife Conservation Society

COGITO AIMS TO:

- Serve as a bridge to geospatial practitioners, educators and learners across public, private and nonprofit sectors
- Study and report on demographics of GIS practitioners
Provide a listing of geospatial resources, classes, universities, certifications
- Explore opportunities to serve as a research hub, providing fellowship grants to researchers studying geospatial data and systems

COGITO MEMBERS

- American Geographic Society
- BetaNYC, Code for America
- Columbia, CIESEN
- Columbia, Earth Institute
- Cornell Tech, The Foundry
- CUNY Mapping Service
- ESRI
- Fund for the City of New York
- GeoNYC Meetup
- GeoDev Northeast Meetup
- GISMO
- Hunter College, CARSI
- National Geospatial-Intelligence Agency
- New York State GIS Association
- NYU Data Services
- Open GeoPortal
- OpenStreetMaps
- Pratt Institute, SAVI
- Pratt Center for Community Development
- The New School
- URISA

VISION



Shankbone, David. (30 October 2012). Flooded Avenue C at East 6th Street in Manhattan's East Village neighborhood of Loisaída, moments before the Con Edison power substation on 14th Street and Avenue C blew up. Retrieved from https://commons.wikimedia.org/wiki/File:Hurricane_Sandy_Flooding_Avenue_C_2012.JPG

THANK YOU!

Check out these great resources:

Ready New York:
<http://www.nyc.gov/readynewyork>

NYC Big Apps:
<http://www.bigapps.nyc>

100 Resilient Cities:
<http://www.100resilientcities.org>

User: King of Hearts. (November 28, 2014). Four-segment panorama of Lower Manhattan, New York City, as viewed from Exchange Place, Jersey City, New Jersey. Retrieve from https://commons.wikimedia.org/wiki/File:Lower_Manhattan_from_Jersey_City_November_2014_panorama_1.jpg